Do Film Production Incentives Attract International Production? A Gravity Model Approach

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Abstract

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1 Introduction

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2 Literature Review

This section reviews relevant literature. Section **??** outlines the global film industry, the increasing use of schemes globally to attract film productions, and the previous literature on the effectiveness of these schemes. Section **??** introduces the gravity model of trade and reviews its application to international services trade.

2.1 International Film Industry

The international film industry was worth an estimated $112 billion in 2025 (**?**), and has also become increasingly globalized in recent decades with the rise of "runaway productions", whereby films originating in one country are physically produced in another either in whole or in part (**?**). A variety of factors are understood to influence the distribution of international film production including labour costs, exchange rates, government incentives, co-production treaties, and the existence of pre-existing production networks (**?????**).

In recent decades, the perceived economic value of hosting film productions such as direct employment and tourism spillovers, as well as the cultural prestige of having an international film industry, has motivated a growing number of national and subnational governments to introduce schemes designed to attract internationally mobile production activity (**?**). These incentive schemes take different forms including rebates, grants, refundable tax credits, and transferable tax credits (**?**). Incentives also vary substantially across jurisdictions with respect to credit rates, caps, eligibility criteria, disbursement timelines, and cultural content requirements (**?**).

The net economic impact of production incentives is debatable. The fiscal cost of film production subsidies often outstrips the tax revenue they generate (**???**). Furthermore, several studies have found evidence of "crowding out" where film production incentives

merely draw resources from other sectors of the economy (?) or otherwise weak economic impacts (??). Conversely, many industry-commissioned studies have found substantial positive returns from such schemes (??). However, the methodological basis of these studies has been questioned, particularly their reliance on elevated multiplier assumptions and counterfactuals that assume no production would have occurred absent the incentive (?).

A related but limited set of literature sets aside the issue of net welfare effects and examines whether incentives succeed in their proximate objective of attracting film productions. Notably, no study in this literature has modelled the determinants of film production location at the international level and instead focuses exclusively on the effectiveness of incentives at the U.S. state level. This is due to the large number of state-level schemes available, as well as the fact that states face an identical federal and international regulatory environment which reduces potential confounding factors. However, this focus on U.S. states means that many potential determinants of production location such as exchange rates and language do not vary and so their impact cannot be quantified.

Some of these studies find meaningful evidence that incentive schemes can attract film productions. ? employs synthetic control and difference-in-differences methods to evaluate the programmes in Louisiana and New Mexico, two early adopters of generous state film incentives, and finds that incentives lead to a statistically significant increase in the number of films produced. ? exploited the fact that California's Film and Television Tax Credit Program allocates tax incentives by lottery, creating quasi-random variation in which productions receive incentives and allowing him to estimate causal effects more cleanly than other studies. He finds that the offer of an incentive meaningfully increased the number of films produced in the state.

Other studies are more mixed. ? estimate a discrete choice model of filming location decisions for productions released between 1999 and 2013. They find significant heterogeneity in responsiveness: mid-sized studios respond to all forms of incentives, major studios respond only to refundable and transferable tax credits, and independent productions exhibit no statistically significant sensitivity to incentives. Finally, ? uses a

panel difference-in-differences design comparing U.S. states before and after they adopt film incentives against similar states that did not adopt over the same period. He finds an increase in the production of TV series but no meaningful effect on feature film production.

2.2 The Gravity Model of Trade

The gravity model of trade predicts that bilateral trade between two countries is proportional to the product of their economic sizes and inversely related to the geographic distance between them (**?**). It has been shown to be derivable from a wide range of trade models including Ricardian (**?**), monopolistic competition (**?**), and heterogeneous firm frameworks (**?**), and has been extensively used to model trade flows (**?**).

When estimating gravity models of international trade, it has become standard to include dummy variables for each exporting and importing country in the sample to control for all country-specific factors that affect a country's overall propensity to export or import. This is because bilateral trade depends on the cost of trade between two countries as well as two multilateral resistance terms (one for the exporter and one for the importer) which capture each country's average trade resistance against all partners. These factors are not directly observable, but because they vary only at the individual country level, they can be absorbed by exporter and importer fixed effects in estimation (**?**).

The gravity model has been extensively applied to goods trade but employed less frequently to trade in services. **?** attribute this to three factors: well-known limitations in bilateral services trade data, the comparatively recent emergence of services as a significant component of international trade, and a widespread assumption that insights from the goods trade literature transfer directly to services. However, existing literature suggests that the gravity framework is well suited to modelling services flows. **?** estimate gravity equations for bilateral services trade between ten OECD member countries and a broader sample of trading partners. Their principal finding is that services trade is better predicted by the gravity specification than goods trade, and that there exists a

complementary relationship between goods exports and services imports.

? applies the gravity model to aggregate services trade and finds that the gravity model fits services trade in a manner broadly comparable to goods. However, he finds that GDP per capita and common language emerge as the most significant determinants of services trade, while geographic distance is generally insignificant. This is consistent with the observation that many services do not require physical shipment and are therefore less sensitive to transport costs but notably diverges from the goods gravity literature which finds distance to be the dominant friction in international trade. **?**, analysing Vietnam–EU bilateral services flows, confirms these patterns and documents the significance of GDP and common language as determinants.

3 Data

3.1 Bilateral Trade Flow Construction

The primary dataset used for this analysis is sourced from the Internet Movie Database (IMDb) and contains the 600 top-grossing films per year from 1990–2023, yielding approximately 20,400 films. Each film contains metadata such as countries of origin, primary filming location, production companies, budget, and release year. Filming locations are extracted by parsing IMDb address strings, taking the final comma-separated element as the country name (e.g., "671 Lincoln Avenue, Winnetka, Illinois, USA" yields "United States"). A comprehensive mapping file harmonises over 160 variant country names and sub-national entities to standardized ISO 3166 country codes. Only films containing valid countries of origin and a filming location which are both different are retained, thereby excluding purely domestic productions. This leaves approximately 7,579 films.

For each film with a valid filming country and at least one identified country of origin, I construct bilateral flows from each origin country to the filming country, excluding domestic pairs where the origin and filming country are identical. Two dependent variables are constructed at the country-pair-year level. The primary dependent variable is the number of films produced per year, while the secondary dependent variable is the total

production budgets in constant 2010 USD, which is available for approximately 47% of films. For films with multiple countries of origin, each origin is paired with this single filming country to generate a separate bilateral flow. When a film has multiple countries of origin, the full budget is divided equally across origin countries.

As a robustness check, I also estimate the model using OECD–WTO Balanced Trade in Services (BaTIS) data from 2005–2023 for "Audiovisual and related services" which captures bilateral exports of film, television, radio, and related production services. This dataset is produced jointly by the OECD and WTO and while limited in the number of observations, provides a consistent and balanced matrix of bilateral trade in services flows. As with the film budgets, these values are deflated to constant 2010 USD.

3.2 Gravity Variables

Standard bilateral gravity variables are sourced from the CEPII GeoDist database and include bilateral distance (population-weighted, in logs), contiguity, common official language, colonial relationship post-1945, and regional trade agreements. Where the CEPII data coverage ends at 2020, the most recent available year is forward-filled to 2023 under the assumption that these predominantly time-invariant or slow-moving variables change minimally over this extension period.

The Economic Freedom of the World (EFW) index data is sourced from the Fraser Institute, both measured for importer and exporter. The EFW index is published annually only from 2000 onward; prior to 2000, it is published at five-year intervals. To obtain a complete annual panel for the 1990–2018 period, linear interpolation is applied within each country's time series for the missing years between 1990 and 2000, as is standard in the literature using EFW data.

Annual GDP data is sourced from the World Bank and is deflated to constant 2015 US dollars, with the natural logarithm of GDP being used in the gravity specification. Inflation data for the purposes of deflating film budgets is sourced from the World Bank.

3.3 Film Incentive Data

Data on national film production incentive schemes for 63 countries is sourced from the Olsberg SPI Global Incentives Index 2025. For each country, the dataset records the year of incentive introduction, the headline rebate or credit rate (as a percentage of qualifying production expenditure), and the incentive type. Incentive types are classified into four categories based on the mechanism through which benefits are delivered to foreign producers: refundable tax credits (where the producer receives cash back even without local tax liability), transferable tax credits (which can be sold to third parties), standard non-refundable tax credits (which offset only local tax liability), and cash rebates (direct cash payments as a percentage of qualifying expenditure). These categories are not mutually exclusive, and two countries have two schemes and are flagged in both applicable categories. Where the Olsberg data reports tiered or variable headline rates, the average across tiers is used.

4 Empirical Methodology

4.1 The Gravity Model Framework

Following **?**, the standard gravity equation drawn from theory takes the following form:

$$T_{ij} = G_i^{\beta_1} \cdot G_j^{\beta_2} \cdot D_{ij}^{\beta_3} \cdot R_i^{\beta_4} \cdot R_j^{\beta_5} \cdot E_{ij} \tag{1}$$

where T_{ij} denotes bilateral trade flows between country i and country j; G_i and G_j represent the economic mass of each country (proxied by GDP); D_{ij} is the geographical distance between countries i and j; R_i and R_j capture the relative distance (remoteness) of each country from the rest of the world; and E_{ij} is a multiplicative error term. Taking logarithms yields the standard log-linear estimating equation.

4.2 Model Specifications

Ten models are estimated, building sequentially from a baseline gravity specification through incentive effects and identification strategies. Models 1–7 are estimated by Poisson Pseudo Maximum Likelihood (PPML) while Models 8–10 (which estimate country-pair effects) are estimated by Ordinary Least Squares (OLS). Models 8–10 are estimated by OLS as the computational demands of country-pair fixed effects mean that estimation by PPML is infeasible.

All PPML models include importer, exporter, and year fixed effects throughout. The fixed effects structure uses separate (non-interacted) importer and exporter dummies rather than the country×year approach of **?**, because the incentive variables of interest vary at the country-year level and would be fully absorbed by country×year fixed effects. This trade-off is discussed further in Section **??**.

4.2.1 Model 1: Baseline Gravity

The baseline specification includes the core gravity variables with importer, exporter, and year fixed effects. Following the PPML framework, the conditional mean is specified in multiplicative form:

$$\mathbb{E}[X_{ijt}] = \exp\left(\beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} + \beta_3 \ln DIST_{ij} + \beta_4 REM_{it} + \beta_5 REM_{jt} + \alpha_i + \alpha_j + \delta_t\right) \tag{2}$$

where X_{ijt} is the level of bilateral flows (trade value, film count, or budget), GDP_{it} and GDP_{jt} are real GDP in constant 2015 US dollars (World Bank), $DIST_{ij}$ is population-weighted bilateral distance from the CEPII Gravity database (**?**), and REM_{it} is a remoteness index constructed following **?** as:

$$REM_{it} = \ln\left(\frac{1}{\sum_{k\neq i}\left[GDP_{kt}/GDP_{Wt}\right]/DIST_{ik}}\right) \tag{3}$$

This index is time-varying because GDP shares evolve, even though distances are fixed. Higher values indicate greater geographic isolation from economically significant partners.

The terms α_i, α_j, and δ_t denote importer, exporter, and year fixed effects respectively, and are included in all PPML specifications (Models 1–7).

4.2.2 Model 2: Full Gravity with Cultural and Institutional Controls

Model 2 augments the baseline with cultural proximity, trade policy, and institutional quality variables inside the exponential:

$$\mathbb{E}[X_{ijt}] = \exp\big(\ldots + \beta_6\, CONTIG_{ij} + \beta_7\, COMLANG_{ij} + \beta_8\, COL45_{ij} + \beta_9\, RTA_{ijt} + \beta_{10}\, EFW_{it} + \beta_{11}\, EFW_{jt}\big) \quad (4)$$

where $CONTIG_{ij}$ indicates a shared land border, $COMLANG_{ij}$ a common official language, $COL45_{ij}$ a post-1945 colonial relationship (all from CEPII), RTA_{ijt} a regional trade agreement in force, and EFW_{it} the Economic Freedom of the World composite index. Common language is expected to be particularly important for audiovisual services, where linguistic accessibility directly affects co-production feasibility and market access. The EFW index is published quinquennially prior to 2000 and annually thereafter.

4.2.3 Model 3: Incentive Dummies

Model 3 introduces the primary variables of interest:

$$\mathbb{E}[X_{ijt}] = \exp\big(\ldots + \gamma_1\, INCENTIVE_{it}^{exp} + \gamma_2\, INCENTIVE_{jt}^{imp}\big) \quad (5)$$

where $INCENTIVE_{it}^{exp}$ equals one if the exporter (filming location) has an active incentive scheme in year t, and $INCENTIVE_{jt}^{imp}$ equals one if the importer (home country) has an active scheme. The coefficient γ_1 captures whether a filming destination's incentive scheme is associated with higher inward production flows. The coefficient γ_2 captures whether the introduction of incentives in a production's home country is associated with increased overseas filming via an "importer incentive" effect whereby domestic incentive infrastructure lowers the fixed costs of organising international production.

4.2.4 Model 4: Incentive Type Heterogeneity

Model 4 disaggregates the exporter incentive into four instrument types (refundable tax credits, transferable tax credits, non-refundable tax credits, and cash rebates):

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_R \, REFUND_{it} + \gamma_T \, TRANSFER_{it} + \gamma_S \, NONREFUND_{it} + \gamma_C \, CASHREBAT \right. \tag{6}$$

These replace rather than supplement the aggregate exporter incentive dummy, avoiding collinearity.

4.2.5 Model 5: Incentive Generosity

Model 5 replaces the binary incentive dummies with continuous generosity measures on both sides:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_1 \, GENEROSITY_{it}^{exp} + \gamma_2 \, GENEROSITY_{jt}^{imp}\right) \tag{7}$$

where $GENEROSITY_{it}$ is the headline rate of the incentive scheme (as a proportion, e.g., 0.25 for a 25% credit), set to zero for countries without an active scheme. This tests whether more generous incentives generate proportionally larger effects on both the exporter and importer sides, sourced from the Olsberg SPI Global Incentives Index.

4.2.6 Models 6 and 7: Cluster Effects

Models 6 and 7 test whether incentive effects strengthen over time through infrastructure accumulation and industry clustering.

Model 6 augments Model 3 with a continuous duration variable:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_1 \, INCENTIVE_{it}^{exp} + \gamma_A \, YEARS_ACTIVE_{it} + \gamma_2 \, INCENTIVE_{jt}^{imp}\right) \tag{8}$$

where $YEARS_ACTIVE_{it} = \min(t - T_i^*, 20)$ when the incentive is active (T_i^* being the introduction year), capped at 20 to prevent extreme values. A positive γ_A would indicate that each additional year of operation generates further production flows, consistent with

learning-by-doing and reputation effects.

Model 7 replaces the aggregate exporter incentive with separate indicators for early and late adopters:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_E\, EARLY_{it} + \gamma_L\, LATE_{it} + \gamma_2\, INCENTIVE_{jt}^{imp}\right) \tag{9}$$

where $EARLY_{it} = 1$ if country i introduced its incentive before 2005 and it is active in year t, and $LATE_{it} = 1$ for post-2005 adopters. If cluster effects matter, early adopters should exhibit a larger coefficient ($\gamma_E > \gamma_L$).

4.2.7 Model 8: Country-Pair Fixed Effects

Model 8 provides the most demanding identification test using country-pair fixed effects, estimated via OLS with a within-transformation. The pair fixed effect α_{ij} absorbs all time-invariant bilateral heterogeneity so only time-varying regressors survive:

$$\begin{aligned} \ln X_{ijt} &= \beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} + \beta_3\, REM_{it} + \beta_4\, REM_{jt} + \beta_5\, RTA_{ijt} \\ &+ \beta_6\, EFW_{it} + \beta_7\, EFW_{jt} + \gamma_1\, INCENTIVE_{it}^{exp} + \gamma_2\, INCENTIVE_{jt}^{imp} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \end{aligned} \tag{10}$$

Estimation proceeds by demeaning all variables by their within-pair means. Singleton pairs (appearing in only one year) are dropped. This specification identifies incentive effects exclusively from within-pair temporal variation: changes in a given pair's flows before versus after incentive introduction.

4.2.8 Model 9: Event Study

Model 9 replaces the binary exporter incentive with relative-time dummies to examine dynamic treatment effects and test for pre-existing trends:

$$\ln X_{ijt} = \boldsymbol{\beta}'\mathbf{z}_{ijt} + \gamma_2\, INCENTIVE_{jt}^{imp} + \sum_{\substack{\tau=-5 \\ \tau\neq-1}}^{+10} \mu_\tau\, D_{i\tau} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \tag{11}$$

where $D_{i\tau}$ equals one if the observation falls τ years relative to the exporter's incentive introduction date, with $\tau = -1$ as the omitted reference. The event window spans $\tau \in [-5, +10]$. The vector $\mathbf{z}_{ijt}$ includes all time-varying controls from Model 8 except the exporter incentive dummy (which is replaced by the event dummies), plus the importer incentive control. Pre-treatment coefficients ($\tau < -1$) test the parallel trends assumption; post-treatment coefficients ($\tau \geq 0$) reveal whether effects materialise immediately, build gradually, or dissipate.

4.2.9 Model 10: Cluster Effects with Pair Fixed Effects

Model 10 combines the years-active variable from Model 6 with the pair fixed effects identification from Model 8:

$$\ln X_{ijt} = \boldsymbol{\beta}'\mathbf{z}_{ijt} + \gamma_1 INCENTIVE_{it}^{exp} + \gamma_A YEARS_ACTIVE_{it} + \gamma_2 INCENTIVE_{jt}^{imp} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \tag{12}$$

The vector $\mathbf{z}_{ijt}$ includes all time-varying controls from Model 8 including the exporter and importer incentive dummies. This tests whether the within-pair incentive effect grows with duration, providing the strictest identification of cluster dynamics by netting out all time-invariant pair heterogeneity.

4.3 Estimator Choice and Fixed Effects Structure

Models 1–7 are estimated by Poisson Pseudo-Maximum Likelihood (PPML), following **?**. PPML models the conditional mean as $\mathbb{E}[X|\mathbf{z}] = \exp(\boldsymbol{\beta}'\mathbf{z})$, estimating directly on the level of the dependent variable. This offers three advantages: it accommodates the count nature of the IMDb dependent variable; it is robust to the heteroskedasticity that invalidates log-linearised OLS; and it naturally handles zero trade flows. Coefficients on log-transformed regressors (GDP, distance) are elasticities; coefficients on dummy variables (incentives, contiguity) are semi-elasticities with percentage effects given by $(\exp(\beta) - 1) \times 100\%$. Model fit is assessed via the McFadden pseudo-$R^2 = 1 - D/D_0$.

All PPML models include separate importer (α_i), exporter (α_j), and year (δ_t) fixed

effects. This controls for all time-invariant country characteristics and common temporal shocks, while allowing time-varying country-level variables (GDP, remoteness, EFW, incentives) to be identified from within-country temporal variation. The theoretically preferred country×year fixed effects of **?** would fully absorb these time-varying unilateral variables, including the incentive dummies that are this paper's primary variables of interest. The separate-FE approach is standard in applied gravity work where the policy variable is unilateral rather than bilateral (e.g., **?**). Models 8–10 are estimated by OLS with country-pair fixed effects implemented via within-transformation (demeaning by pair means). Year dummies are included.

Throughout all specifications, standard errors are clustered at the country-pair level. In panel trade data, residuals for a given country pair are likely correlated across years: if a pair experiences an unusually productive year for film co-production, the following year is also likely to be above average due to persistent relationship-specific factors such as ongoing co-production agreements. Pair-level clustering produces standard errors that are robust to both heteroskedasticity (unequal error variance across pairs of different sizes) and arbitrary within-pair serial correlation (**??**). This is more conservative than heteroskedasticity-only robust standard errors, which assume independence across observations and correct only for unequal variance.

4.4 Robustness and Subsample Analysis

Each pair fixed effects specification (Models 8, 9, and 10) is estimated on four samples to address potential threats to identification.

4.4.1 The Anglo-Network Problem

The English-speaking production network (the United States, United Kingdom, Canada, Australia, and New Zealand) dominates global production flows and may introduce counterintuitive distance coefficients, since these geographically dispersed countries trade intensely due to shared language, institutional similarities, and deep co-production relationships. Models are re-estimated on two restricted samples: (a) excluding Anglo-to-Anglo

pairs only, and (b) excluding all pairs involving any Anglo country. Stability of incentive coefficients across these subsamples indicates results are not driven by idiosyncratic features of the Anglo network.

4.4.2 Sample Without Economic Freedom Controls

Because the EFW index reduces sample coverage, all pair FE models are re-estimated dropping EFW controls entirely, using the broader merged dataset. This verifies that results are not an artefact of sample selection induced by EFW data availability.

4.4.3 Income-Adjusted Economic Freedom (EFWRESID)

Following **?**, who note the correlation between economic freedom and per capita income, all ten models are additionally estimated replacing the raw EFW index with an income-adjusted variant. The residualised index ($EFWRESID$) is constructed by regressing EFW on the log of GDP in a pooled cross-sectional OLS regression across all country-year observations:

$$EFW_{it} = \alpha + \beta \cdot \ln GDP_{it} + \varepsilon_{it} \tag{13}$$

$$EFWRESID_{it} = \hat{\varepsilon}_{it} \tag{14}$$

The residual captures the component of economic freedom that is orthogonal to income level—institutional quality beyond what a country's GDP would predict. Separate regressions are run for exporter and importer sides. Stability of incentive coefficients across the raw EFW and EFWRESID specifications confirms that results are not confounded by the EFW–income correlation.

4.4.4 Dual Dependent Variables (IMDb)

For the IMDb dataset, all specifications are estimated with two dependent variables: the bilateral film count (extensive margin) and the aggregate bilateral budget in constant 2010 USD (intensive margin). Consistency across these two measures strengthens the

interpretation that incentives affect the real volume of production rather than merely the number of low-budget projects.

4.4.5 Cross-Dataset Validation

The parallel estimation across IMDb (film-level microdata, 1990–2023) and BaTIS (official services trade statistics, 2005–2023) provides independent validation. The two datasets have different units, coverage, and measurement error structures, so convergent results provide stronger evidence than either alone.

Table 1: Summary of Model Specifications

	Description	Est.	Fixed Effects	Dep Var	Identifying Variation
M1	Baseline gravity	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M2	Full gravity (+ cultural, RTA, EFW)	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M3	Incentive dummies	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M4	Incentive type dummies	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M5	Generosity (exp + imp)	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M6	Cluster: years active	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M7	Cluster: early vs late	PPML	Imp, Exp, Year	Levels	Cross-sect. + temporal
M8	Incentive dummies	OLS	Pair, Year	Log	Within-pair temporal
M9	Event study	OLS	Pair, Year	Log	Within-pair, rel. time
M10	Cluster: years active	OLS	Pair, Year	Log	Within-pair temporal

5 Results

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6 Limitations

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7 Conclusion

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Appendix

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